



DEVELOPMENT OF AN AUTOMATIC SPEECH-TO-SPEECH SYNTHESIS ARCHITECTURE FOR ENGLISH-TO-YORÙBÁ LANGUAGE USING NEURAL MACHINE TRANSLATION APPROACH

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Presented By

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OUTLINE

1. Abstract
2. Introduction
3. Literature Reviewed
4. Methodology
5. Conclusion
6. References

1. ABSTRACT

- Different researchers have developed various tools and applications to ensure seamless communication between human beings for day-to-day transactions and interactions.
- Speech-to-Speech synthesis entails translating the speech of one language to the speech of another language.
- This proposal aims to launch the development of speech-to-speech resources using the neural machine translation approach.
- The focus of the work will be translational use case between a medical practitioner and patient as well as public health practitioner and public.
- An extension to Translatotron 2 architecture will be developed in this proposed work to translate from English to Yoruba languages.

▪ To develop the proposed model, Yorùbá-English general domain and medical domain speech corpora will be curated, and two models will be trained and developed.

▪ These are the base model and the transfer learning model using an already pre-trained English language model to another language.

▪ What follows is the training of the two models using the general domain speech corpora of Yorùbá-English curated.

▪ Second is the focus of the work, where the medical domain data will be utilised to train the two models.

▪ Thereafter comparison of the models developed under different experiments will be carried out.

▪ Finally, a prototype Softbot shall be developed and evaluated.

2. INTRODUCTION

2.1 Background

English language is the third most widely spoken native language in the world behind Mandarin Chinese and Spanish, as well as the most spoken language in the world spoken by nearly 1.5 billion speakers, which is slightly higher than 1.1 billion Mandarin Chinese speakers (Silaban et al., 2023).

Yoruba is a language that shares membership with the Niger-Congo language family (Abiola et al., 2014) and one of the three official languages in Nigeria, spoken by a population of about 50 million across South Western Nigeria, coupled with the USA, UK, Brazil, Benin and Togo (Akintola & Ibiyemi, 2017).

The translation of speech from one language to another is one of the research areas in speech synthesis that has gained recognition in recent years (Chang et al., 2023), and different approaches have been utilised to achieve this task.

The approaches are illustrated in Figure 1, while Figures 2 to 5 show the models available for each component of the cascaded system.

- **United Nations SDGs**
 - ✓ Goal 3 – Ensuring healthy lives and well-being
 - ✓ Goal 10 – reducing inequality within and among the community.
- **Industrial Revolution 4(IR4)**

Conventional Approach

- Cascaded ASR+ NLP+MT+SS (Jia et al., 2019; Rubenstein et al., 2023)
- ASR: Automatic Speech Recognition
- NLP: Natural Language Understanding
- MT: Machine Translation
- SS: Speech Synthesis/Text-to-Speech (TTS)

End-to-End Approach

- Deep Learning (Limbu, 2020)
- Encoder/Decoder Network
- Encoder can be network or stacks of MLPs, BiLSTM, Conformers etc.
- Decoder can be network of LSTMs

Figure 1: Approaches to Modelling Speech-to-Speech Synthesis

2. INTRODUCTION

2.1 Background (Cont.)

Well-known S2ST models developed

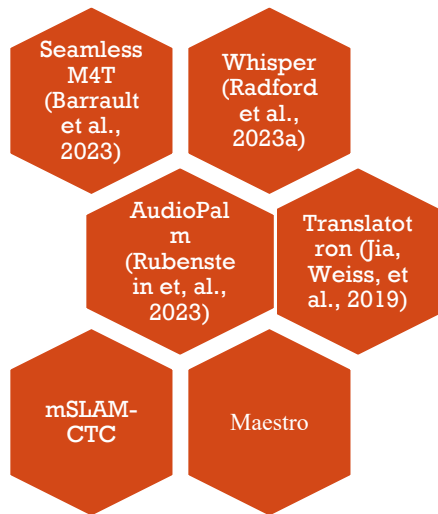


Figure 2: Well-known S2ST Models

Well-known MT models developed

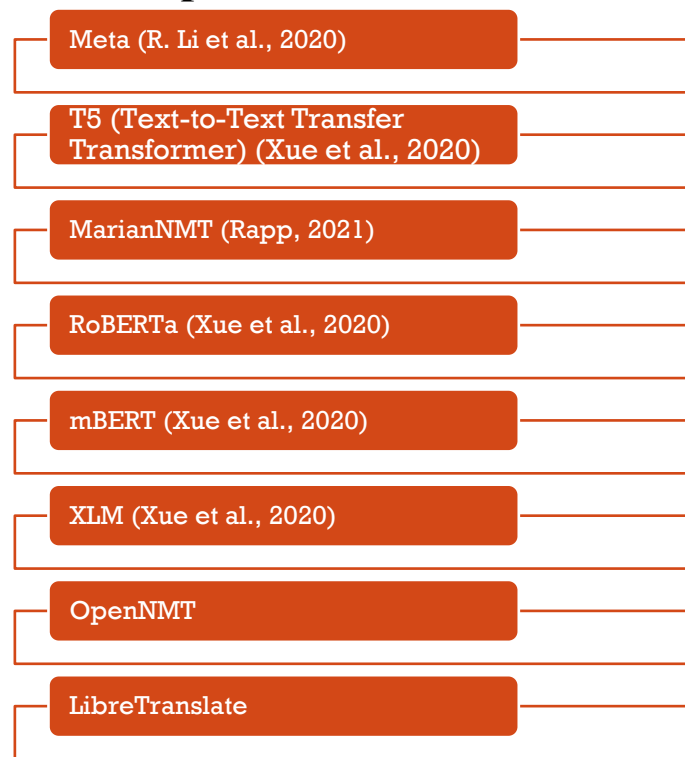


Figure 3: Well-known MT Models

Well-known TTS models developed

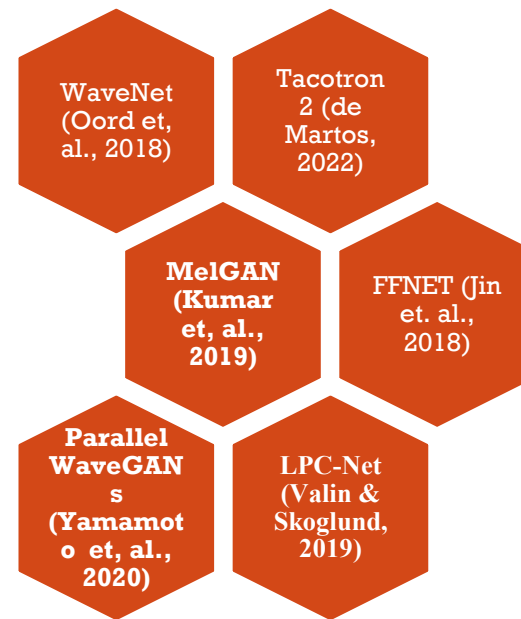


Figure 5: Well-known TTS Models

Well-known ASR models developed

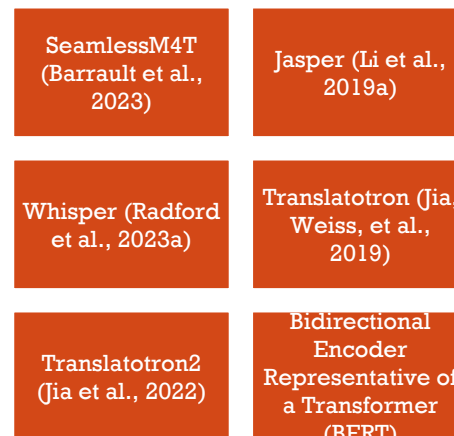


Figure 4: Well-known ASR Models

2. INTRODUCTION

2.2 Problem Statements

- The number of available speech resources or tools that are not sufficient as well as Yorùbá speech corpus. (Bansal, 2019; Zhou et al., 2013, Fagbolu et al., 2015; Meyer et al., 2022) .
- The output audio samples of the target language are not real representation of the input audio samples (Limbu, 2020).
- Medical speech corpora are not available for training (Cherednichenko, 2020; Grabar & Cardon, 2019.; Qian et al., 2023).
- Language barriers (Hudelson & Chappuis, 2024; Javaid et al., 2023; Panayiotou et al., 2020; Tu et al., 2024)
- The uncertain alignment between the input and output spectrograms is challenging (Jia, Weiss, et al., 2019).
- Delays and errors are associated with the cascaded system.

2.3 Justification for the Study

- Speech-to-speech translation can minimize the language barriers experienced by individuals in their conversations with colleagues who speak other languages.
- Direct speech-to-speech translation to be considered in this work has the benefit of preserving the prosody, emotion, accent, and voice of the speaker.
- More speech data for English-Yoruba language pairs will be publicly available.

2. INTRODUCTION

2.4 Aim and Objectives

- The aim of this research is to develop an automatic Speech-to-Speech Synthesis Model for English-to-Yorùbá Language based on Neural Machine Translation Approach.
- The objectives to realise the aim are:

I. create general and public health speech corpus for English-to-Yorùba language pair with focus on the accent of speakers of the two languages.

II. preprocessing of the data collected in steps i, and ii

III. design and implement an improved architecture for speech-to-speech translation (S2ST) of English-to-Yoruba language pair.

IV. develop a prototype Softbot based on the S2ST architecture in (iii) for malaria public health education use case.

V. carry out performance evaluation of the S2ST architecture and the prototype Softbot.

2. INTRODUCTION

2.5 Scope of the Study

- Improved speech-to-speech synthesis model from English-to-Yorùbá language translation will be developed.
- The parallel speech corpora, will be curated and to be made available for research community.
- The corpora will include general and public health domains.
- A prototype Softbot based on the S2ST model will be developed

2.6 Expected Contribution to Knowledge



1. General and public health corpus for English-to-Yoruba direct S2ST corpus will be created and open-sourced for the language technology research community.
2. Improved data preprocessing algorithms for audio corpus
3. An improved direct S2ST architecture for English-to-Yoruba that can be finetuned for other low-resourced African language pairs.
4. A crowd-sourcing web application and a prototype S2ST Softbot that could be used for translation within the public health education domain and adapted for other public address translation use cases.

3. LITERATURE REVIEW

3.1 Generic Architecture of Direct S2ST

❑ The architecture of a direct S2ST is illustrated in Figure 6 (Jia et al., 2019; Jia et al., 2022)

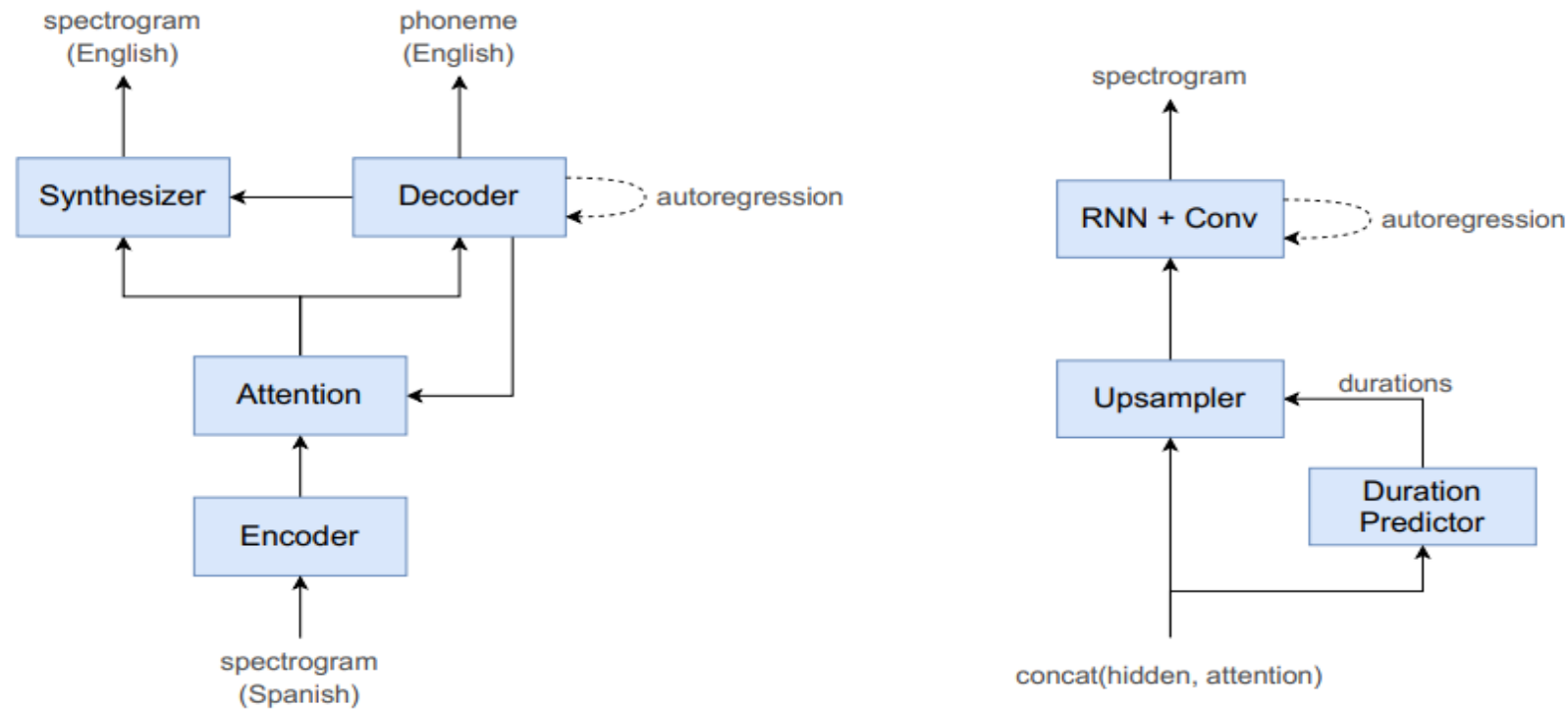


Figure 6: Generic Direct Speech-to-Speech Architecture (Translatotron 2) (Jia et al., 2022)

3. LITERATURE REVIEW

3.2 Transformer Architecture of Direct S2ST

- ❑ Model will also be developed for transformer architecture
is illustrated in Figure 7 (Vaswani et al., 2017)

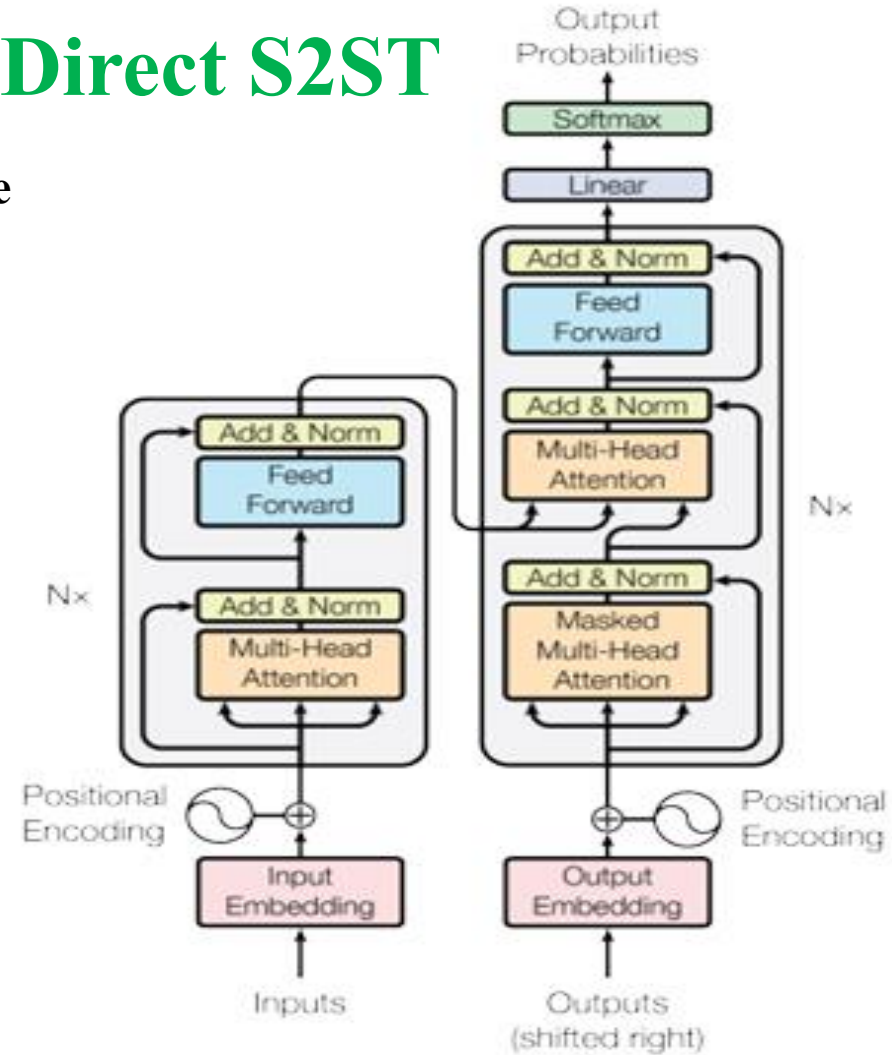


Figure 7: Transformer Architecture (Vaswani et al., 2017)

3. LITERATURE REVIEW

3.3 Summary of Literature Review



Table 1: Review of Related Works

Ref	METHODS	DATASETS/ LANGUAGE	RESULTS	FEAATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Chang et al., 2023)	Sequence-to-Sequence modelling using Convolution Sub-Sampling ESPnet Toolkit for implementation	Dataset: MuST-C Average of 430 hours of audio recordings of TED talks. Languages: English-German, Spanish, French	BLEU scores using FBaank: En-De: 26.7 En-Es: 31.2 En-Fr: 37.1 BLEU scores using Discrete tokens: En-De: 28.6 En-Es: 33.0 En-Fr: 38.7	(1) Use of discretisation for speech extracted features. (2) Use of De-duplication and sub-word modelling to further reduce the size of the speech features.	Advantages 1. Performance enhancement 2. Reduction in training time due to discrete token. Limitations 1. The work did not consider the comparison of other discretisation approach

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/L LANGUAGE	RESULTS	FEAATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Jia et al., 2019)	Attention Sequence-to-Sequence approach with: Speaker Encoder, and Neural Vocoder Translatotron	Datasets: Conversational Spanish-to-English dataset: 979k parallel utterance pairs Fisher Spanish-to-English dataset: Spanish-English	Conversational Spanish-to-English dataset: BLEU scores: Source + Target model: 42.7, Cascaded model (STT & TTS): 48.7, Ground truth: 74.7 Fisher Spanish-to-English	Mel-spectrogram (MFCCs)	Advantages 1. It explore different model trainings. 2. It utilises multi-head attention for speech-to-speech translation. Limitation 1. The computational time of the algorithm was not considered.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/LANGUAGES	RESULTS	FEATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Jia et al., 2022)	Direct speech-to-speech Translatotron 2	Datasets: Conversational es→en, Fisher es→ en, Multilingual CoVoST2 (es, fr, de, ca→en) Languages: Spanish-English, French-English, German-English, Catalan-English	Translation Quality Fisher dataset: outperformed Translatotron by 15.5 BLEU score. Subjective MOS Naturalness Fisher dataset: Translatotron: 3.70 ± 0.08 Translatotron2: 3.98 ± 0.08	Mel-spectrogram features	Problems it addresses from the original Translatotron: 1. Non-contribution of the attention alignment learned. Advantages 1. Performance was closer to the cascaded system, Limitations 1. it does not consider the discrete rep.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/L LANGUAGE	RESULTS	FEAATURE EXTRACTI ON METHODS	ADVANTAGES / LIMITATIONS
(Julius et. al., 2025)	Direct speech-to-speech (Transformer architecture) Augmented Attention Mechanism (73 hours / 71 hours)	Datasets: 200, 023 audio recordings Source: Primary Public domain Language: English-Yoruba	BLEU SCORE: 0.7, MOS: 2.8, Translation Accuracy: 98 Original Transformer translation Accuracy: 88.16%	Spectrogram based model	Advantages: 1. Augmented attention mechanism to capture speech representation such as the diacritics, prosody, intonation etc. 2. More attention head (42) utilized for training. Limitations: 1. The dataset could be scaled up for enhanced performance. 2. training input data is spectrogram based. 3. A trade-off between choice of attention-head and complexity.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/L LANGUAGE	RESULTS	FEAATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Prabhjot et. Al., 2024)	Sequence-to-sequence model: Direct Unit-to-Unit Translation (U2UT) model	Datasets: 1. FLEURS with 1625 parallel Punjabi-English speech. 2. Synthetically generated English speech Language: Punjabi-English	U2UT Model: BLEU score: 3.9829 S2UT Model: BLEU score: 0.2954 Cascaded Model (Whisper-large-v3): BLEU score: 18.1136	U2UT Model: No feature extraction was utilised. HuBERT Base model trained on 960 hours of Librispeech	Advantages: 1. The use of discrete speech units that reduces the dimension of the speech signal while also preserving the original Limitations: 1. The size of the dataset was not sufficient enough for training; hence performance of the proposed approach was lower than the cascaded system implemented with Whisper-large-v3.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/LANGUAGE	RESULTS	FEATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Radford et al., 2023)	Transformer encoder & decoder using MLP AdamW optimiser	LibriSpeech, TED-LIUM 3, Common Voice 5.1, Artie bias Corpus, Call Home & Switch board, WSJ, CORAAL, CHiME-6, AMI-IHM AMI-SDM1, Meanwhile, Rev16, Kincaid 46 Earnings-21, etc. LANGUAGE: Non specific, multilingual ASR	Transcription Multilingual LibriSpeech: Whisper large-v2 (Spanish): WER = 4.2 %, Common Voice 9: Whisper large-v2 Dutch: WER = 5.8 % Spanish: WER = 5.6 % FLEURS:	GELU activation function, MFCC	ADVANTAGES 1. Robust to noise (additive noise & pub noise) LIMITATIONS 1. Ideal robustness could not be achieved. 2. Performance varied across different dataset.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/L LANGUAGE	RESULTS	FEAATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Rubenstein et al., 2023)	Discretisation + Decoder-only transformer model Models: AudioPaLM 8B AST, AudioPaLM 8B S2ST, AudioPaLM-2 8B AST,	Datasets: CoVoST2, CVSS, VoxPopuli, Common Voice, etc. Language: Nonspecific	CoVost2 Dataset (AST Task): BLEU scores: , AudioPaLM-2 8B AST: 37.8, AudioPaLM-2 8B cascaded ASR + translation: 39.0 Whisper Large-v2: 29.1	No feature extraction Discrete tokenisation	Advantages: 1. It performed better than the SOTA Whisper-Large-v2 Model. Limitations: 1. It is not a sequence-to-sequence modelling, 2. There might be possible delays introduced during training.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/L LANGUAGE	RESULTS	FEAATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Olatunji et al., 2023)	Collection of texts / transcripts from sites such as PubMed, NCBI, wiki-texts, developme nt of audio recording web page.	Datasets: AfriSpeech Language: English language of African accent from 13 countries.	AfriSpeech-clinical data Whisper medium: WER of 0.568 Wave2Vec large: WER of 0.415 LibriSpeech(test-clean) Whisper large pre-trained model: WER of 0.167	Nil	Advantages: 1. Accented corpus. 2. Curated for general and medical domain. Limitations: 1. Not curated for African indigenous languages.

3. LITERATURE REVIEW (CONT.)

3.3 Summary of Literature Review

Table 1: Review of Related Works (Cont.)

Ref	METHODS	DATASETS/L LANGUAGE	RESULTS	FEAATURE EXTRACTION METHODS	ADVANTAGES / LIMITATIONS
(Anmol Gulati et al., 2020)	Conformer model for ASR.	Datasets: LibriSpeech Language: English language.	Big Conformer model with 118 million model parameters without a LM: WER = 0.021 With LM: WER = 0.019	Spectrogram-based	Advantages: 1. Ability to capture long term dependencies (global) by transformer and local relationship by th help of CNN Limitations 1. Just LSTM decoder was utilized for the end-to-end ASR model developed.

3. LITERATURE REVIEW (CONT.)

3.4 Gaps in the Literature

Table 2: Summary of Gaps in the Literature

Consideration	Authors	Gaps
Dataset	(Bansal, 2019); (Kaur et al., 2024); (Cherednichenko, 2020); (Grabar & Cardon, 2019); (Qian et al., 2023); (Kaur et al., 2024)	Insufficient amount of medical and nonmedical parallel speech corpus for the modelling of speech-to-speech translation Difficulty in acquiring parallel speech corpus
Complexity of the architectures	(Jia et al., 2022)	The system is complex due to the architecture implemented, and the computational complexity of the models is not mostly stated

3.4 Gaps in Literature

Table 2: Summary of Gaps in Literature (Cont.)

Consideration	Authors	Gaps
Speech discretisation	(Chang et al., 2023) (Rubenstein, Asawaroengchai, Nguyen, Bapna, Borsos, Quitry, Chen, Badawy, Han, Kharitonov, & others, 2023).	Limitation on work done Possible generation of errors due to lack of discretisation
Cross-lingual voice transfer	(Jia, Weiss, et al., 2019); (Kaur et al., 2024)	Challenges are encountered with cross-language voice transfer, and performance was lesser than the cascaded system

3. LITERATURE REVIEW (CONT.)

3.4 Gaps in Literature

Table 2: Summary of Gaps in Literature (Cont.)

Consideration	Authors	Gaps
Complexity of the architectures	(Jia et al., 2022)	The system is complex due to the architecture implemented, and the computational complexity of the models is not mostly stated

4. METHODOLOGY

Table 3: Table of Methodology to Achieve Specific Objectives

Specific Objectives	Methodology	Data	Expected Results
1	State-of-the-art text-to-text translation tools, Recording of the parallel corporal of both languages using a user friendly web site, Use of SPSS/MATLAB for analysis.	English text and corresponding Yorùbá text, Crowdsourced Human beings.	Quality accented English-Yorùbá parallel corpus
2	Sampling, Feature extraction: MFCCs, Discretisation: Hubert Modelling, K-means clustering,	Outputs from Objective 1	Discretised features with reduced dimension ready to be fed as input to the model.

4. METHODOLOGY

Table 3: Table of Methodology to Achieve Specific Objectives

Specific Objectives	Methodology	Data	Expected Results
3	Sequence-to-modelling using transformer architecture, and training	Outputs from Objective 2	State-of-the-art English-Yorùbá language translation model with
4	Utilisation of various software packages like the Python, HTML,CSS, APIs etc.	Output from Objective 3	User friendly Softbots language translator.
5	Computation of BLEU scores, and MOS Naturalness and MOS Similarity, as well as testing of Softbot by selected individual	Outputs from Objective 3 and 4	SOTA BLEU scores and MOS that is typical of a S2ST models.

4. METHODOLOGY

4.1 Proposed Architecture

- Figure 8 is the architecture of this proposed study to provide a broader understanding of the work.
- An improvement on Translatotron 2 (Figure 6)

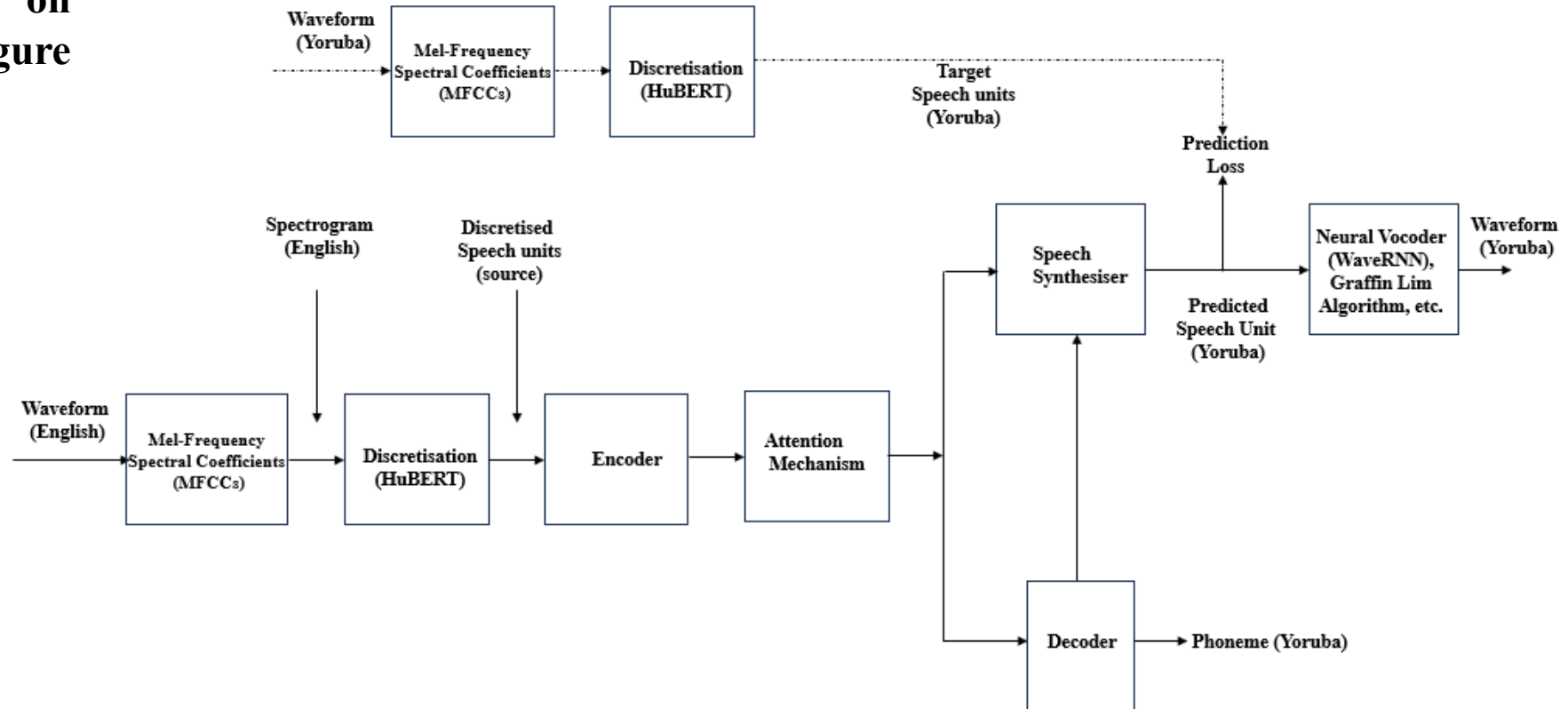


Figure 8:The Proposed Speech-To-Speech Neural Machine Translation Architecture

4. METHODOLOGY

4.1 Proposed Architecture (Cont.)



- The speech synthesizer of Figure 8 is shown in Figure 9.
- It is an extension to Translatotron 2, a SOTA speech-to-speech model.

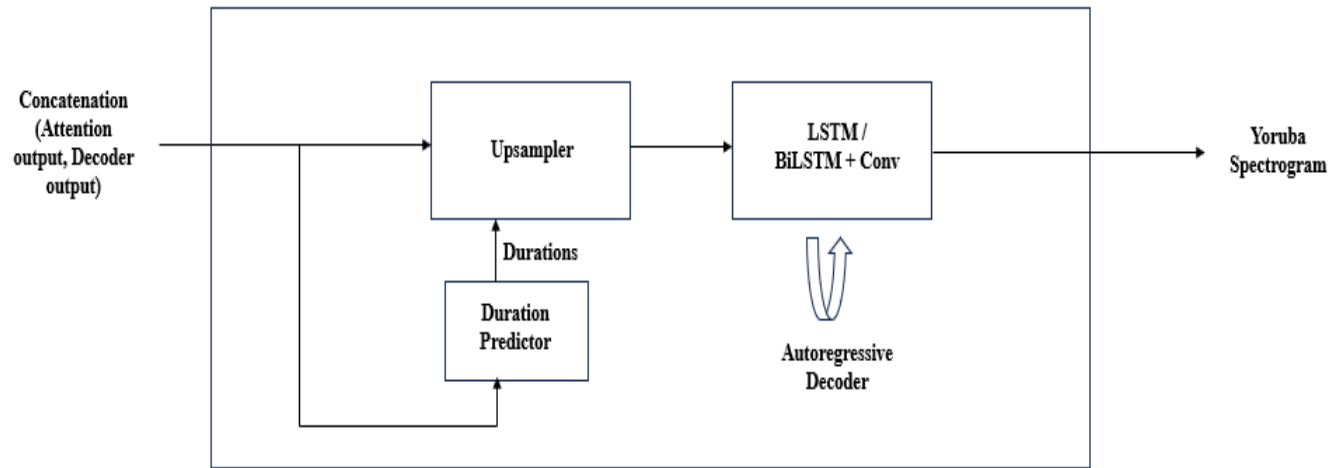


Figure 9:The Speech Synthesizer of Figure 8

3.1.1 Major Extension to be carried out

- English -Yoruba language pair corporal to be utilised for training.
- Speech discretization to be utilised unlike MFCC features, for both source and target languages.
- Stacks of LSTM, BiLSTMs shall be consider for the encoder and decoder.
- LSTM, BiLSTM shall be considered for the decoder of the speech synthesizer along with convolutional network.



4. METHODOLOGY

4.1 Proposed Architecture (Cont.)



Table 4: Translatotron2 versus Proposed Architecture

S/N	Varying Blocks	Translatotron2	Proposed Architecture
1	Training data nature	Spectrogram based	discretised
2	Encoder	Stacks of Conformer blocks: -Feed-forward layer -self-attention layer -convolutional layer -second feed-forward layer	-Same encoder as used in a transformer architecture (stack) (Vaswani et al., 2017), also to experiment with conformer block -Stack of LSTM
3	Linguistic Decoder	Stack of LSTM	Stack of LSTM/BLSTM/GRU
4	Synthesizer	Stack of LSTM	Stack of LSTM/BLSTM/GRU
5	Source / Target language	Spanish / English	English / Yoruba

4. METHODOLOGY

4.1 Proposed Architecture (Cont.)

4.1.1 Encoder

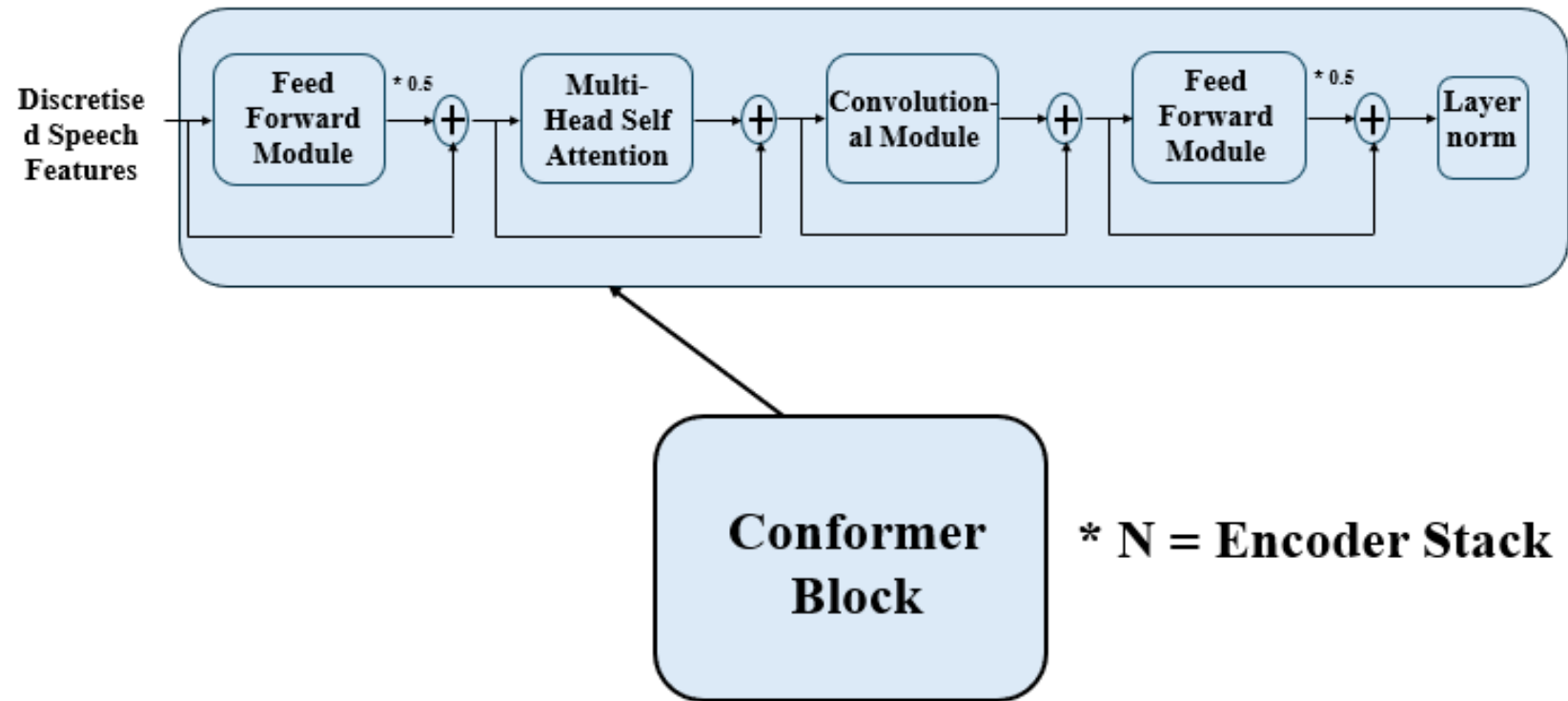


Figure 10: Conformer Block

4. METHODOLOGY

4.1 Proposed Architecture (Cont.)

4.1.1 Encoder

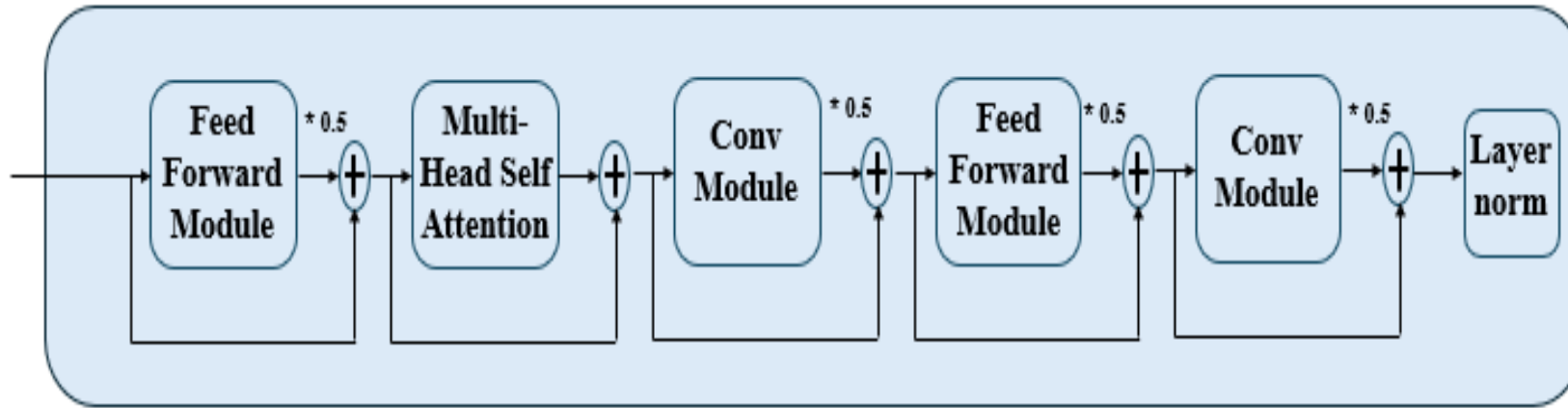


Figure 11: Conformer Block with Altered Convolutional Module

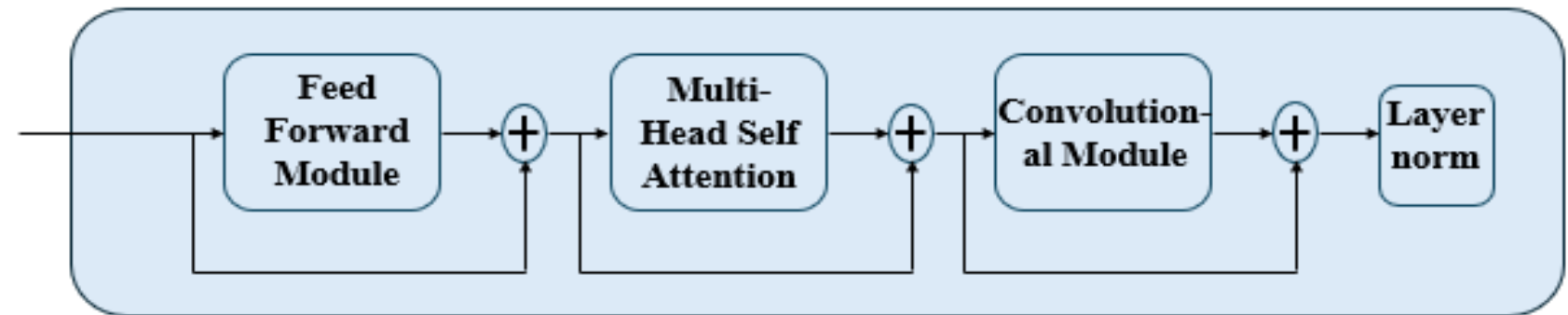


Figure 12: Conformer Block with Altered Feed Forward Module

4. METHODOLOGY

4.1 Proposed Architecture (Cont.)

4.1.1 Encoder

A. Feed Forward Module

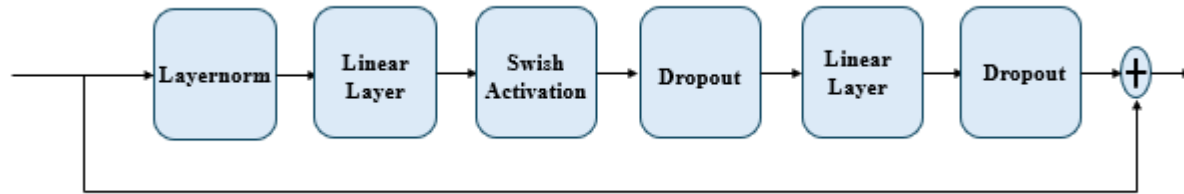


Figure 13: Feed Forward Module

B. Multi-Head Self Attention (MHSA) Module

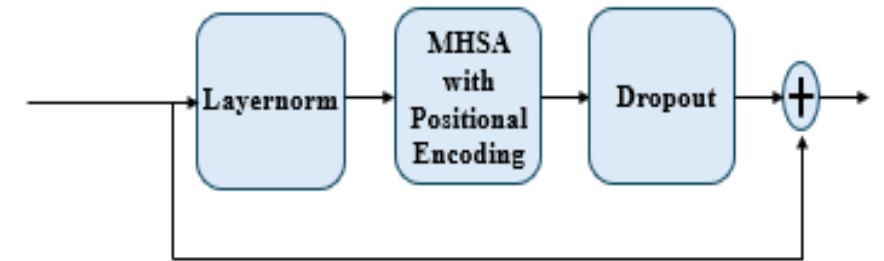


Figure 15: Multi-Head Self Attention Module

C. Convolutional Module

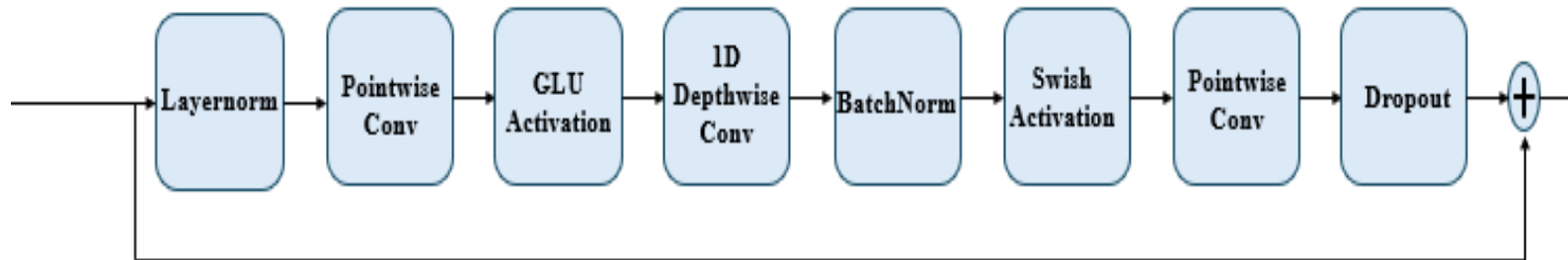


Figure 14: Convolutional Module

4. METHODOLOGY

4.1 Proposed Architecture (Cont.)

4.1.1 Encoder

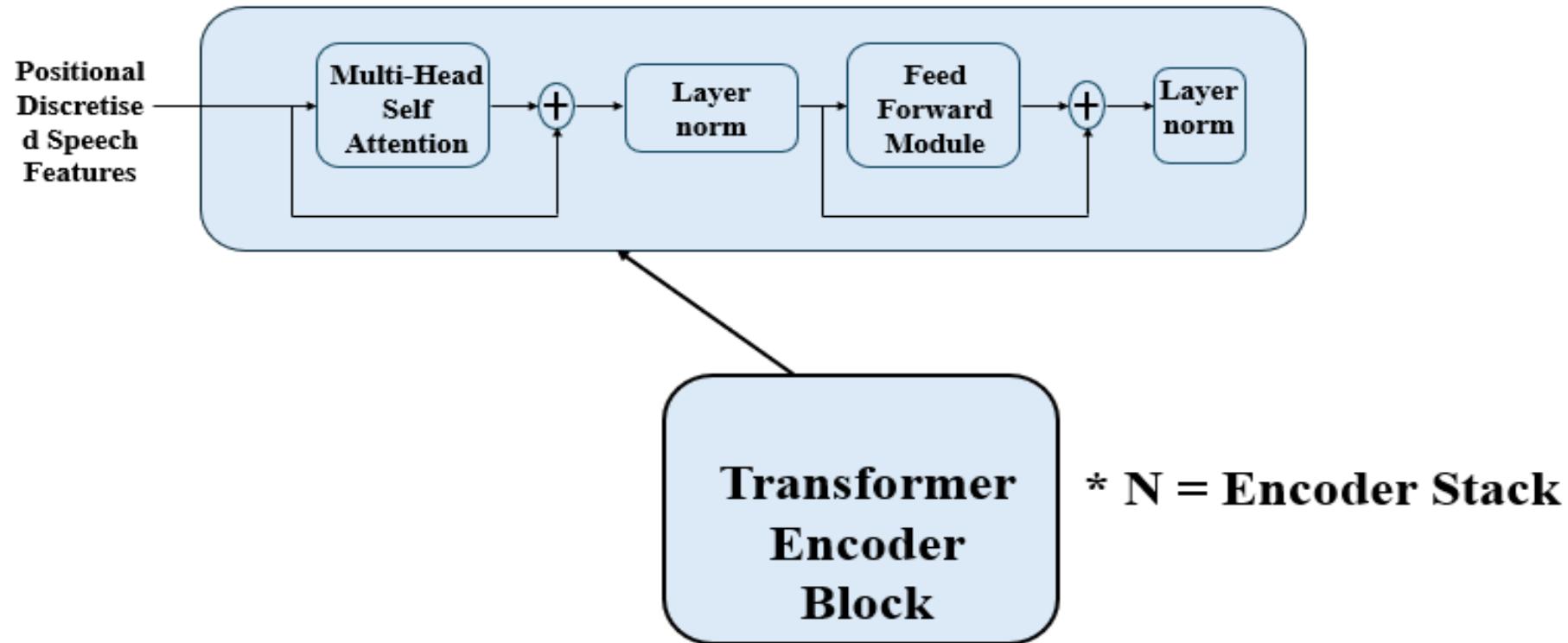
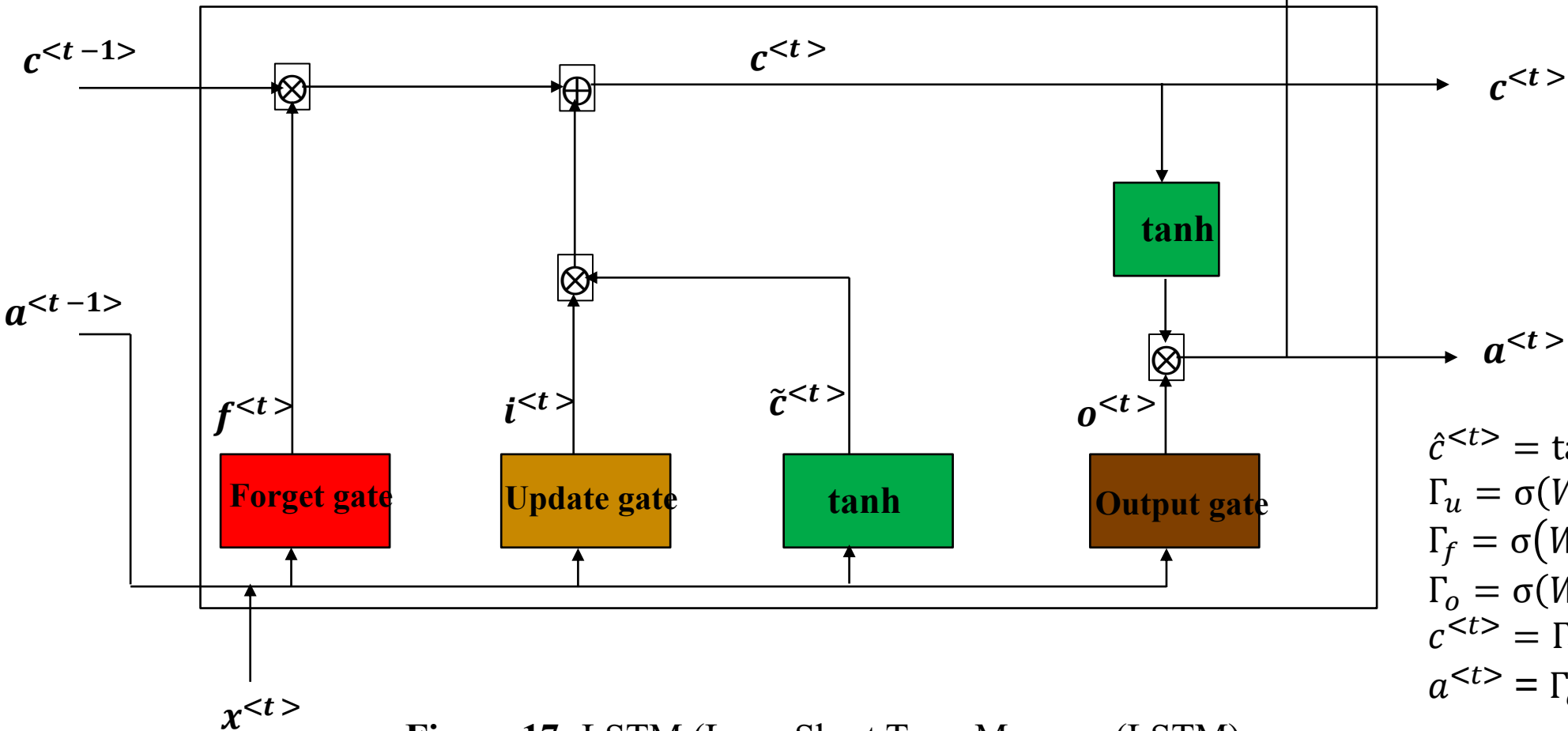


Figure 16: Transformer Encoder

4. METHODOLOGY

4.1 Proposed Architecture (Cont.)

4.1.2 Decoder (Stack of LSTMs)



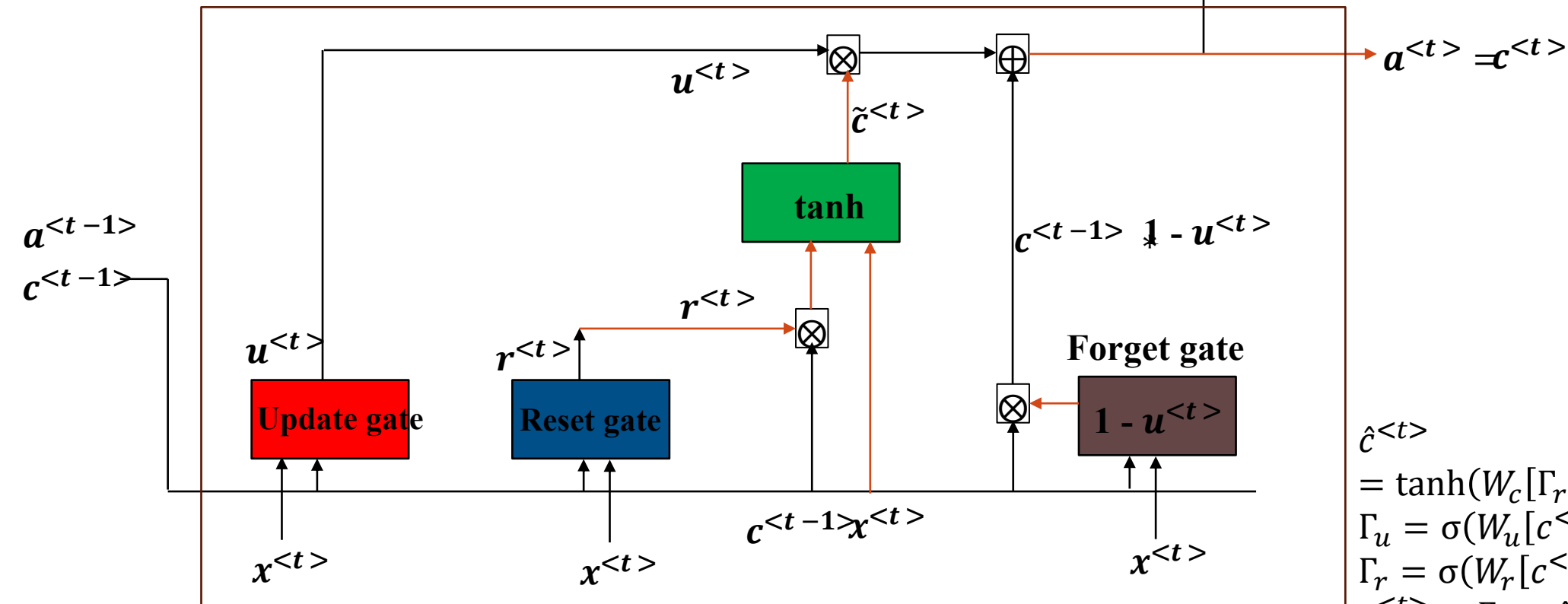
$$\begin{aligned}\hat{c}^{<t>} &= \tanh(W_c[a^{<t-1>}, x^{<t>}] + b_c) \\ \Gamma_u &= \sigma(W_u[a^{<t-1>}, x^{<t>}] + b_u) \\ \Gamma_f &= \sigma(W_f[a^{<t-1>}, x^{<t>}] + b_f) \\ \Gamma_o &= \sigma(W_o[a^{<t-1>}, x^{<t>}] + b_o) \\ c^{<t>} &= \Gamma_u * \hat{c}^{<t>} + \Gamma_f * c^{<t-1>} \\ a^{<t>} &= \Gamma_o * \tanh c^{<t>}\end{aligned}$$

Figure 17: LSTM (Long Short Term Memory (LSTM))

4. METHODOLOGY

4.1 Proposed Architecture (Cont.)

4.1.2 Decoder (Stack of GRUs)



$$\begin{aligned}\hat{c}^{<t>} &= \tanh(W_c[\Gamma_r * c^{<t-1>}, x^{<t>}] + b_c) \\ \Gamma_u &= \sigma(W_u[c^{<t-1>}, x^{<t>}] + b_u) \\ \Gamma_r &= \sigma(W_r[c^{<t-1>}, x^{<t>}] + b_r) \\ c^{<t>} &= \Gamma_u * \hat{c}^{<t>} + (1 - \Gamma_u) * c^{<t-1>} \\ a^{<t>} &= c^{<t>}\end{aligned}$$

Figure 18: LSTM (Gated Recurrent Unit (GRU))

4.1 Proposed Architecture (Cont.)

4.1.2 Decoder (Transformer Decoder)

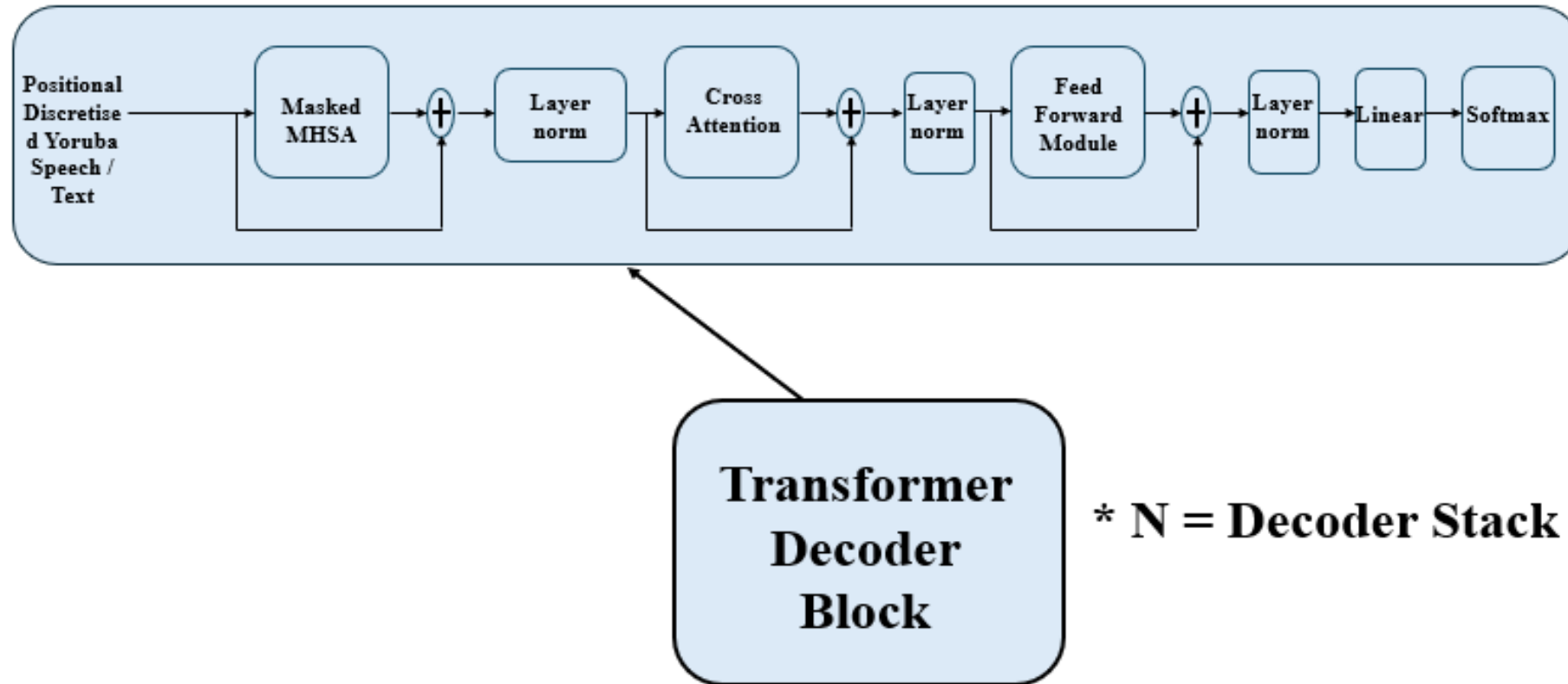


Figure 19: Transformer Decoder

4. METHODOLOGY

4.2 Flow Chart of the Proposed Model

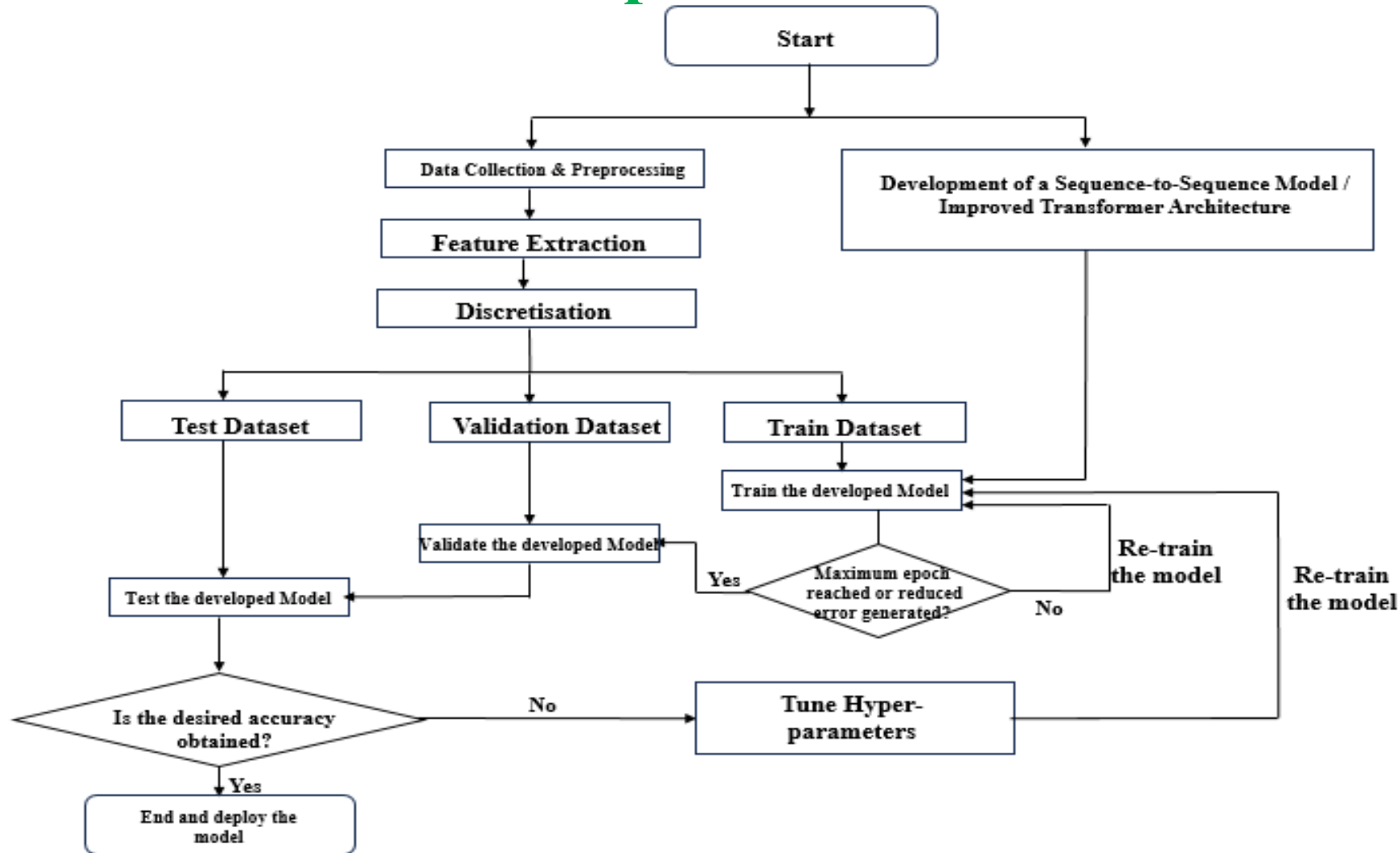


Figure 20: The Flowchart of the proposed Model

4.3 Outcomes of Method 1: Data Curation Feedback

4.3.1 Overall Statistics (03/03/2025 – 15/03/2026)

Table 5: Overall Statistics [186,424 audios (409.82 hrs.); Speakers (470 + 8); # 5,994,124.9 – 15/03/2026]

S/N	Institutions	No of participants Projected	No of Registered participants	No of participants that completed	Amount Spent
1	UNILAG	100	Whatsapp – 50; Online - 38	22 (PH: 11; GEN: 11) + 5	# 434,124.90
2	UI	100	Whatsapp – 159; Online – 143, New Whatsapp – 194	84 (PH: 50; GEN: 34) + 2	# 860,000.00
3	FUNAAB	100	Whatsapp – (99+40); Online – 59	45 (PH: 20; GEN: 25)	# 450,000.00
4	LASU	100	Whatsapp – 90; Online - 88	5 (PH: 1; GEN: 4)	# 50,000.00
5	OAU	100	Whatsapp – None; Online – None	None	None
6	UI_2 + UI_3	100	Whatsapp – 200+; Online – not yet verified	96 (PH: 57; GEN: 39) 99 (PH: 96; GEN: 3)	# 960,000.00 # 990,000.00
7	UI_BATCH2	100	Whatsapp – 180;	106 (PH: 45; GEN: 61)	# 1,060,000.00
8	UI_BATCH2_2	100	Whatsapp – 180;	13 (PH: 12; GEN: 01)	# 130,000.00

4.3 Outcomes of Method 1: Data Curation Feedback

4.3.2 University of Lagos

Table 6: UNILAG Statistics (22 + 5 Speakers)

1	num_audios_src	4253
2	num_audios_tgt	4253
3	total_num_audios (src+tgt)	8506
4	len_audio(time)_src (secs)	28961.746 (8 hrs.)
5	len_audio(time)_tgt (secs)	29151.519 (8 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	58113.26 (16 hrs.)
7	num_words_src	40404
8	num_unique_words_src*	16825
9	num_words_tgt	44247
10	num_unique_words_tgt*	16098
11	num_of_characters_src	193862
12	num_of_characters_tgt	171099
13	num_lines_of_text_src	4253
14	num_lines_of_text_tgt	4253
15	num_lines_of_text_total (src + tgt)	8506

4.3 Outcomes of Method 1: Data Curation Feedback

4.3.3 University of Ibadan, UI

Table 7: UI Statistics (84 + 2 Speakers)

1	num_audios_src	16,545
2	num_audios_tgt	16,545
3	total_num_audios (src+tgt)	33090
4	len_audio(time)_src (secs)	114240.87 (31.73 hrs.)
5	len_audio(time)_tgt (secs)	127335.19 (35.37 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	241576.08 (67.10 hrs.)
7	num_words_src	164230
8	num_unique_words_src*	64501
9	num_words_tgt	178084
10	num_unique_words_tgt*	64869
11	num_of_characters_src	781462
12	num_of_characters_tgt	686416
13	num_lines_of_text_src	16,545
14	num_lines_of_text_tgt	16,545
15	num_lines_of_text_total (src + tgt)	33090

4.3 Outcomes of Method 1: Data Curation Feedback

4.3.3 University of Ibadan, UI (Cont.)

Table 8: UI 2 Statistics (96 Speakers)

1	num_audios_src	19,089
2	num_audios_tgt	19,089
3	total_num_audios (src+tgt)	38,178
4	len_audio(time)_src (secs)	112,449.6225 (31.24 hrs.)
5	len_audio(time)_tgt (secs)	129,585.15547 (36.00 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	242,034.77797 (67.24 hrs.)
7	num_words_src	160,992
8	num_unique_words_src*	54,432
9	num_words_tgt	178,327
10	num_unique_words_tgt*	47,971
11	num_of_characters_src	899,072
12	num_of_characters_tgt	785,255
13	num_lines_of_text_src	19,089
14	num_lines_of_text_tgt	19,089
15	num_lines_of_text_total (src + tgt)	38,178

4.3 Outcomes of Method 1: Data Curation Feedback

4.3.3 University of Ibadan, UI (Cont.)

Table 9: UI 3 Statistics (99 Speakers)

1	num_audios_src	19,670
2	num_audios_tgt	19,670
3	total_num_audios (src+tgt)	39,340
4	len_audio(time)_src (secs)	143,350.773567 secs (39.82 hrs.)
5	len_audio(time)_tgt (secs)	171,491.200867 secs (47.64 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	314,841.958434 secs (87.46 hrs.)
7	num_words_src	223,063
8	num_unique_words_src*	70,973
9	num_words_tgt	2261,647
10	num_unique_words_tgt*	63,274
11	num_of_characters_src	1,293,412
12	num_of_characters_tgt	1,182,595
13	num_lines_of_text_src	19,670
14	num_lines_of_text_tgt	19,670
15	num_lines_of_text_total (src + tgt)	39,340

4.3 Outcomes of Method 1: Data Curation Feedback

4.3.3 University of Ibadan, UI (Cont.)

Table 10: UI_BATCH2 Statistics (106 Speakers)

1	num_audios_src	21,125
2	num_audios_tgt	21,125
3	total_num_audios (src+tgt)	42,250
4	len_audio(time)_src (secs)	173,129.24764 secs (48.09 hrs.)
5	len_audio(time)_tgt (secs)	212,023.47656 secs (58.90 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	385,152.7242 secs (106.99 hrs.)
7	num_words_src	305,759
8	num_unique_words_src*	114,019
9	num_words_tgt	355,668
10	num_unique_words_tgt*	118,356
11	num_of_characters_src	1,531,102
12	num_of_characters_tgt	1,473,213
13	num_lines_of_text_src	21,125
14	num_lines_of_text_tgt	21,125
15	num_lines_of_text_total (src + tgt)	42,250

4.3 Outcomes of Method 1: Data Curation Feedback

4.3.3 University of Ibadan, UI (Cont.)

Table 11: UI_BATCH2_2 Statistics (13 Speakers)

1	num_audios_src	2,596
2	num_audios_tgt	2,596
3	total_num_audios (src+tgt)	5,192
4	len_audio(time)_src (secs)	21,009.01674 secs (05.84 hrs.)
5	len_audio(time)_tgt (secs)	25,866.04205 secs (07.19 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	46,875.0588 secs (13.03 hrs.)
7	num_words_src	29,611
8	num_unique_words_src*	11,757
9	num_words_tgt	36,531
10	num_unique_words_tgt*	11,764
11	num_of_characters_src	171,640
12	num_of_characters_tgt	172,575
13	num_lines_of_text_src	2,596
14	num_lines_of_text_tgt	2,596
15	num_lines_of_text_total (src + tgt)	5,192

4.3 Outcomes of Method 1: Data Curation Feedback

4.3.4 Federal University of Agriculture Abeokuta, FUNAAB

Table 12: FUNAAB Statistics (45 Speakers)

1	num_audios_src	8,953
2	num_audios_tgt	8,953
3	total_num_audios (src+tgt)	17,906
4	len_audio(time)_src (secs)	78,831.28686 (21.90 hrs)
5	len_audio(time)_tgt (secs)	89,138.5185 (24.76 hrs)
6	len_audio(time)_total (src + tgt) (secs)	167,969.80536 (46.66 hrs)
7	num_words_src	102,197
8	num_unique_words_src*	36,157
9	num_words_tgt	111,779
10	num_unique_words_tgt*	37,566
11	num_of_characters_src	500,223
12	num_of_characters_tgt	445,647
13	num_lines_of_text_src	8,953
14	num_lines_of_text_tgt	8,953
15	num_lines_of_text_total (src + tgt)	17,906

4.3 Outcomes of Method 1: Data Curation Feedback

4.3.5 Lagos State University, LASU

Table 13: LASU Statistics (5 Speakers)

1	num_audios_src	981
2	num_audios_tgt	981
3	total_num_audios (src+tgt)	1,962
4	len_audio(time)_src (secs)	8,633.685334 (2.39 hrs.)
5	len_audio(time)_tgt (secs)	10,137.173334 (2.81 hrs.)
6	len_audio(time)_total (src + tgt) (secs)	18,770.858664 (5.21 hrs.)
7	num_words_src	12,373
8	num_unique_words_src*	4,691
9	num_words_tgt	13,016
10	num_unique_words_tgt*	5,012
11	num_of_characters_src	56,754
12	num_of_characters_tgt	50,552
13	num_lines_of_text_src	981
14	num_lines_of_text_tgt	981
15	num_lines_of_text_total (src + tgt)	1,962

4.3 Outcomes of Method 1: Data Curation Feedback

4.3.6 Percentage Collected

Table 14: PERCENTAGE OF NUMBER OF AUDIOS COLLECTED

NUMBER OF AUDIOS PROJECTED = 200,000; OBTAINED PER LANGUAGE: 93,212 AUDIOS

	UNILAG	UI	FUNAAB	LASU	UI_2	UI_3	UI_Batch2
Total number of EN	4,253 /18,181.81 (23.39%)	16,545 /18,181.81 (91.00%)	8,953/18,181.81 (49.24%)	981/18,181.81 (5.40%)	19,089/18,181.81 (104.99%)	19,670/18,181.81 (108.19%)	21,125/18,181.81 (116.19%)
Total number of YOR	4,253 /18,181.81 (23.39%)	16,545 /18,181.81 (91.00%)	8,953/18,181.81 (49.24%)	981/18,181.81 (5.40%)	19,089/18,181.81 (104.99%)	19,670/18,181.81 (108.19%)	21,125/18,181.81 (116.19%)
Overall %	$4253 + 16545 + 8,953 + 981 + 19,089 + 19,670 + 21,125 + 2,596 = 93,212 / 200,000 = 46.61\%$						

4.3 Outcomes of Method 1: Data Curation Feedback

4.3.6 Percentage Collected

Table 14: PERCENTAGE OF NUMBER OF AUDIOS COLLECTED (CONT.)

NUMBER OF AUDIOS PROJECTED = 200,000; OBTAINED PER LANGUAGE: 93,212 AUDIOS

	UI_Batch2_2						
Total number of EN	2,596/18,181.81 (14.28%)						
Total number of YOR	2,596/18,181.81 (14.28%)						
Overall %	$4253 + 16545 + 8,953 + 981 + 19,089 + 19,670 + 21,125 + 2,596 = 93,212 / 200,000 = 46.61\%$						

4.3.6 Percentage Collected (Contd)

Table 15: PERCENTAGE OF HOURS OF AUDIOS COLLECTED

HOURS OF AUDIOS PROJECTED = 1,000 hrs.;

OBTAINED: 189.05 hrs. (EN), 220.77 hrs. (YOR), TOTAL = 409.82 hrs.

	UNILAG	UI	FUNAAB	LASU	UI_2	UI_3	UI_Batch 2
Total hours of EN	8.04	31.73	21.90	2.39	31.24	39.82	48.09
Total hours of YOR	8.10	35.37	24.76	2.81	36.00	47.64	58.90
Total hours (Expected = 90.90 hrs. (9.09%))	16.14 (1.61%)	67.10 (6.71%)	46.66 (4.67%)	5.21 (0.52%)	67.24 (6.72%)	87.46 (8.75%)	106.99 (10.70%)
Overall %	$16.14 + 67.10 + 46.66 + 5.21 + 67.24 + 87.46 + 106.99 + 13.03$ $= 409.82 / 1,000 = 40.98 \%$						

4.3.6 Percentage Collected (Contd)

Table 15: PERCENTAGE OF HOURS OF AUDIOS COLLECTED (CONT.)

HOURS OF AUDIOS PROJECTED = 1,000 hrs.;
OBTAINED: 189.05 hrs. (EN), 220.77 hrs. (YOR), TOTAL = 409.82 hrs.

	UI_Batch2_2						
Total hours of EN	05.84						
Total hours of YOR	07.19						
Total hours (Expected = 90.90 hrs. (9.09%))	13.03 (1.30%)						
Overall %	$16.14 + 67.10 + 46.66 + 5.21 + 67.24 + 87.46 + 106.99 + 13.03 = 409.82 / 1,000 = 40.98 \%$						

FROM 22 + 84 + 45 + 5 + 96 + 99 + 106 + 13 = 470 DOCUMENTED SPEAKERS / 1100 OBTAINED; % OF SPEAKERS = 42.73 %

4. METHODOLOGY (CONT.)

4.4 Budget

Table 16: Research Budget

S/n	ITEM	COST (\$)	QUANTITY	TOTAL (\$)
1	Speech Data Curation (Google PhD)		10 weeks	10,064.14
3	Conferences (CApIC ACE)	750	2	1,500
4	Publications (CApIC ACE)	2000	2	4,000
	Total			15,564.14

4.5 Timelines

Table 17: Research Timeline Detailing Semester Breakdown of Activities

TASK NO	TASK NAME	START DATE	END DATE	OMEGA SEMESTER , 2023/2024	ALPHA SEMESTER 2024/2025	OMEGA SEMESTER 2024/2025	ALPHA SEMESTER 2025/2026	OMEGA SEMESTER 2025/2026
1	Development of Proposal	04/01/2024	01/06/2024	Yes				
2	Field Work	09/04/2025	30/03/2026		Yes	Yes	Yes	
3	Methodology	09/04/2025	30/03/2026			Yes	Yes	
4	Results and Discussion	01/01/2026	30/03/2026				Yes	
5	Completion of Report	01/04/2026	27/08/2026					Yes

Table 18: Field Work Breakdown

APRIL-JUNE, 2025	JULY-SEPTEMBER, 2025	OCTOBER, 2025	NOVEMBER-DECEMBER, 2025	JANUARY, 2026	FEBRUARY, 2026	MARCH, 2026
Generation / Curation of dataset Analysis of dataset	Generation / Curation of dataset Understanding of datasets / installation of software packages, modules Analysis of dataset	Generation / Curation of dataset Analysis of dataset	Generation / Curation of dataset Analysis of dataset	Analysis of dataset	Analysis of dataset	Performance analysis of the developed models: Base and Transfer Learning models
Sending out of notification message to intended individual for the task	Familiarisation with the installed software, and collected parallel audio files, as well as dataset available Knowledge of Python programming for deep learning Creation of manifest files	Familiarisation with python modules for the aforementioned models Knowledge of Python programming for deep learning Creation of manifest files	Preparation of dataset for public access Statistical analysis Creation of manifest files	Preparation of dataset for public access Statistical analysis Creation of manifest files	Compare the transfer learning results obtained with that of the BASE MODEL developed. Preparation of dataset for public access Statistical analysis Creation of manifest files	Editing and proof reading of the whole of chapter 3 for corrections, and update Preparation of dataset for public access Statistical analysis Creation of manifest files
Development of web site for collection of data through crowdsourcing	Crowdsourcing and recording in UNILAG, LASU, FUNAAB, and UI	Crowdsourcing and recording in OAU, FUTA, University of Ado-Ekiti, and UNILORIN	Crowdsourcing and recording in University of Lome, Togo, and Univ. of Abomey Calavi, Benin Republic	Crowdsourcing and recording in CU		
Sending out of links for the collection of audio data to intended individual.	Thorough analysis of the dataset for deep understanding, and interpretation using matlab, spss, possibly R	Knowledge of discritisation like the HuBERT, Wav2Vec models and how they are implemented				.

Table 19: Methodology Breakdown

APRIL-JUNE, 2025	JULY-SEPTEMBER, 2025	OCTOBER, 2025	NOVEMBER-DECEMBER, 2025	JANUARY, 2026	FEBRUARY, 2026	MARCH, 2026
Generation / Curation of dataset Development of website Scraping online websites for texts transcripts Use of text-to-text translation Crowd-sourcing individuals for recording	Generation / Curation of dataset Development of models for data preprocessing Development of models for discretization, encoder, speaker encoder, attention mechanism, decoder, and vocoder	Generation / Curation of dataset Development of models for data preprocessing Development of models for discretization, encoder, speaker encoder, attention mechanism, decoder, and vocoder	Generation / Curation of dataset Development of models for data preprocessing Development of models for discretization, encoder, speaker encoder, attention mechanism, decoder, and vocoder	Integration of the developed models into a single base model Train, Validate, Test, and carryout performance evaluation Develop Transfer Learning model Train, Validate, Test, and carryout performance evaluation	Train, Validate, Test, and carryout performance evaluation Develop a prototype Softbot	Integration of developed models to the prototype softbot
Use of Python Flask with HTML, CSS and Vercel server for recording webpage. Thorough search for online media database, newspapers, etc. Use of Google translate Crowdsource individual from 11 institutions to participate in recordings	Same as previous Analysis of data Familiarizations with Python modules for the packages to implement the models. Development of the models mostly taking into consideration the discretization with HuBERT, transformer models.	Same as January Knowledge of the choice of the hyperparameters, for various models of the expected base model. Gain further insight from literatures	Same as February Test each of the developed models. Make a choice of the language translation pretrained model to be utilised for transfer learning	Split the preprocessed discretised dataset into train, validate and test, train the training data using the single developed base, and transfer learning models evaluate using metrics like the MOS, BLEU, scores, precision, recall, F1-score, and ROUGE. [using the FEDGEN platform to run the code]	Train the training data using the single developed base and transfer learning model, evaluate using metrics like the MOS, BLEU, scores, precision, recall, F1-score, and ROUGE. [using the FEDGEN platform to run the code] – For both medical and non medical corpus Ablation studies	Development of another website (Softbot) for interfacing with users. Use of APIs to integrate the model output to the developed web site interface. Sourcing of raters to access the performance of the Softbot

Table 20: Results and Discussion Breakdown

JANUARY, 2026	JANUARY, 2026	FEBRUARY, 2026	FEBRUARY, 2026	MARCH, 2026	MARCH, 2026
Analysis of the results of the various experimental setup for transfer learning and base model for medical and non medical speech corpus	Analysis of the results of the various experimental setup for transfer learning and base model for medical and non medical speech corpus	Updating of the experimental parameters and extraction of new results	Development of write up for the discussion	Development of write up for the discussion	Thorough look through the developed chapter contents for experimental errors and possible corrections.
I shall do both analytical illustration of the results. And graphical illustration if needed.	I shall do both analytical illustration of the results. And graphical illustration if needed.	Based on the already obtained results, there might need to update or enhance the performance of the models via change of hyperparameters of the model.	A coherent discussion of the obtained results will be carried out without bias.	A coherent discussion of the obtained results will be carried out without bias.	Editing and proof reading Checking for experimental hyperparameters erroneously utilised for each of the experiments.
Use of MATLAB, Python for analysis, and interpretation of results	Use of MATLAB, Python for analysis, and interpretation of results	I shall adjust these parameters and report possible improvement if any or as a limitation of the study when certain parameters are adjusted.	Discussion of the findings (analysis, tables, possible chat) will be reported and the whole document of chapter 4 will be integrated as a single file.	Discussion of the findings (analysis, tables, possible chat) will be reported and the whole document of chapter 4 will be integrated as a single file.	Drawing the summary of the chapter.

Table 21: Completion of Report

APRIL - JUNE, 2026	JULY-AUGUST, 2026
Merging of all chapters into a single document	Submission of report
Merge the proof read chapter 1, chapter 2, chapter 3, chapter 4, chapter 5, and appendix into a single document	Submission of the completed report for Viva
	Corrections of errors pointed out.

4.5.5 Summary of Timeline

Table 22: Summary of Research Timeline

S/N	ACTIVITY	YEAR ONE												YEAR TWO												YEAR THREE											
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
1	Defining Research Focus																																				
2	Literature Review																																				
3	Developing Methodology																																				
4	Ethical Application																																				
5	Data Collection																																				
6	Data Analysis																																				
7	Proposal																																				
8	Thesis Writing																																				
9	Publication																																				
10	M I L E S T O N E																																				
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5. CONCLUSION

- **The development of speech-to-speech modelling has seen revolution and improvement for high-resourced languages like English, Spanish, French, and German.**
- **However, the African indigenous languages, including the Yoruba language, are still underdeveloped in language translation from another language to Yoruba and vice versa.**
- **This proposed work investigates how to publicly make available parallel English-Yoruba speech corpora to aid further studies in model development for other researchers.**
- **This work will leverage the transfer learning approach to train medical speech corpus and non-medical speech corpus to obtain a robust speech translation output which will further guarantee the utilisation of speech translation in the medical space, as well as public engagement between medical health practitioners and public thereby addressing the SDGs 3 and 10 and the IR4.**

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